Hemanth Chowdary Kavula

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with **2+ years of experience** developing and maintaining ETL/ELT pipelines using **Python, SQL, PySpark, and AWS**. Hands-on experience with **AWS Glue, Amazon EMR, Amazon S3, Snowflake, dbt, and Apache Spark** for data ingestion, transformation, and analytics-ready datasets. Skilled in **dimensional modeling, SQL optimization, automated data quality validation, and production troubleshooting**, with exposure to Kafka and Kinesis streaming. Experienced in supporting reliable data delivery for **financial reporting, risk analytics, and downstream AI/ML applications**.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Scala, Java
Cloud Platforms: AWS (S3, Glue, EMR, Kinesis, Redshift, Lambda)
Big Data & Streaming: Apache Spark, PySpark, Spark SQL, Apache Kafka, Delta Lake, Databricks
Databases & Data Warehousing: Snowflake, PostgreSQL, MySQL, SQL Server, Oracle
Data Integration & Orchestration: ETL/ELT, dbt, Apache Airflow, Data Ingestion
Data Modeling & Quality: Dimensional Modeling, Star Schema, SCD Type 2, Great Expectations, Data Validation, Data Reconciliation
DevOps & Analytics: Git, Jenkins, Terraform, Docker, Power BI, Tableau
Python Data Libraries: Pandas, Scikit-learn

PROFESSIONAL EXPERIENCE

Fidelity Investments | Jersey City, NJ — Jan 2026 – Present

Data Engineer

- Developed and maintained **ETL/ELT pipelines using Python, SQL, PySpark, AWS Glue, and Amazon EMR** to process financial and trading datasets for downstream analytics and reporting.
- Engineered **PySpark transformations** to cleanse, filter, join, and aggregate raw data stored in **Amazon S3**, producing structured datasets for analytical workloads.
- Supported **near-real-time data ingestion using Apache Kafka and AWS Kinesis**, enabling trading event data delivery to downstream processing and monitoring systems.
- Developed **Snowflake SQL transformations and dbt models** to convert financial source data into structured, analytics-ready datasets supporting reporting and downstream analysis.
- Implemented automated validation checks using **Python, SQL, and Great Expectations** to detect duplicates, missing values, and schema inconsistencies before downstream data loading.
- Prepared and validated **Delta Lake datasets using PySpark**, supporting financial analytics and downstream AI/ML applications, including risk analysis and anomaly detection.
- Monitored **ETL pipeline executions**, investigated processing failures through log analysis and validation checks, and supported code deployments using **Git and Jenkins**.
- Optimized **SQL queries and PySpark transformations** by refining join strategies, applying predicate filtering, and adjusting workflows to improve **ETL processing efficiency**.

Mastercard | San Francisco, CA — May 2025 – Sep 2025

Data Engineer Intern

- Developed **Python and SQL scripts** under engineering team guidance to extract, clean, and transform structured datasets for internal analytics and reporting.
- Assisted with **ETL pipeline development, testing, and data loading**, validating source-to-target transformations across relational databases.
- Developed **SQL queries using JOINs, subqueries, aggregations, and filtering** to support data extraction, reporting, and transformation validation.
- Performed SQL-based source-to-target reconciliation to **identify missing records, duplicates, and data discrepancies**, documenting validation findings for the engineering team.
- Supported **AWS S3 data processing and ETL troubleshooting**, collaborating with engineers to investigate data loading issues and document pipeline workflows.

Arshil Digital Private Limited | Hyderabad, India April 2023 – Nov 2023

Junior Developer Intern

- Assisted in developing and maintaining software applications using **Python and SQL**, supporting application functionality and data processing tasks.
- Developed SQL queries using **joins, subqueries, filtering, and aggregations** to retrieve, update, and validate relational data.
- Created Python scripts for **file processing, data cleaning, and basic validation** to support routine development activities.
- Supported application debugging and unit testing by investigating issues, implementing small code changes, and validating fixes under senior developer guidance.
- Collaborated with developers to document changes, maintain source code using **Git**, and support application development workflows.

PROJECTS

Customer Lifetime Value (CLV) & Churn Prediction | GitHub

- Designed a **MySQL star schema** with transaction fact and customer, product, and date dimension tables to support customer analytics.
- Developed SQL transformations using **CTEs and window functions** for RFM segmentation, customer-level aggregations, and feature engineering.
- Built Python-based CLV and churn models using **BG/NBD, Gamma-Gamma, and Logistic Regression**, generating 6- and 12-month CLV estimates and evaluating classification performance using **ROC-AUC, precision, and recall**.

Technologies: Python, MySQL, SQL, Pandas, Scikit-learn, BG/NBD, Gamma-Gamma

Pizza Sales Analysis | GitHub

- Developed **14 SQL queries** across four normalized MySQL tables using JOINs, CTEs, subqueries, aggregations, and window functions to analyze sales transactions and revenue performance.
- Applied **SUM OVER and DENSE_RANK** to calculate cumulative revenue, identify top-performing pizzas by category, and analyze peak ordering hours and sales trends.

Technologies: MySQL, SQL, CTEs, Window Functions, Relational Data Modeling

Personal Portfolio Website | GitHub | Live Site

- Developed and deployed a responsive portfolio website using **React, Vite, Tailwind CSS, and Netlify** to showcase technical projects, certifications, and professional experience.

Technologies: React.js, JavaScript, Vite, Tailwind CSS, GitHub, Netlify

EDUCATION

Master of Science in Computer Science Jan 2024 – Dec 2025

Rowan University | Glassboro, NJ

PUBLICATIONS

Age and Gender Prediction using Gated Residual Attention Network | IEEE (2023) | View Publication

CERTIFICATIONS

SQL (Advanced) Skill Certification — HackerRank | May 2025 | Verify
AWS Academy Cloud Foundations — AWS Academy | Feb 2022 | Verify
Foundations: Data, Data, Everywhere — Google / Coursera | Nov 2025 | Verify